

# Construction for Strongly $k$ -Hamiltonian Graphs \*

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## Abstract

The  $k$ -Hamiltonian graphs have been studied by many researchers. In this paper, we introduce strongly  $k$ -Hamiltonian graphs. A strongly  $k$ -Hamiltonian graph is also  $k$ -Hamiltonian. Moreover, we present two construction schemes for strongly  $k$ -Hamiltonian graphs including  $(k + 2)$ -join and Cartesian product with  $K_2$ . Applying these schemes, we can construct many new strongly  $k$ -Hamiltonian graphs.

*Keywords:*  $k$ -Hamiltonian, strongly  $k$ -Hamiltonian graphs,  $(k + 2)$ -join, Cartesian product, interconnection network, fault tolerance.

## 1 Introduction

An interconnection network is usually represented by a graph. Let  $G = (V, E)$  be an undirected graph, where  $V$  is the *node set* and  $E$  is the *edge set* of  $G$ . The *degree* of vertex  $v$  in  $G$ , denoted by  $d_G(v)$ , is the number of edges incident to  $v$ . A *path* is a sequence of nodes in which every two consecutive nodes are adjacent. A *Hamiltonian path* of  $G$  is a path  $V$  if every node of  $V$  is exactly visited once. A graph is *Hamiltonian connected* if there exist Hamiltonian paths

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\*This work was supported in part by the National Science Council of the Republic of China under Contract NSC 90-2213-E-212-012.

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between every pair of its nodes. A *cycle* is a path if the first node and the last node are the same node. A *Hamiltonian cycle* is a cycle which traverses every node of  $V$  exactly once. A *Hamiltonian graph* is a graph that contains a Hamiltonian cycle.

Fault tolerance is desirable in the design of interconnection networks. The faults of a network correspond to removing edges and/or nodes from the graph. Let  $F \subseteq (V \cup E)$  denote the faults of graph  $G$ . Let  $G - F$  denote the subgraph induced by  $V - F$  of  $G$  and then deleted the edges in  $F$  from this induced subgraph. A graph  $G$  is a *k-Hamiltonian graph* if  $G - F$  is Hamiltonian for all  $F \subset (V \cup E)$  and  $|F| = k$ . In this paper, we introduce a new concept, called strongly *k-Hamiltonian graphs*, for the fault tolerance of Hamiltonian graphs. A graph  $G$  is a *strongly k-Hamiltonian graph* if  $\forall v \in (V - F)$  there exist  $k + 2 - |F|$  edges incident to  $v$  such that every pair of these edges is on some Hamiltonian cycle of  $G - F$ , for all  $F \subseteq (V \cup E)$  and  $|F| \leq k$ . Thus, a strongly *k-Hamiltonian graph* is a *k-Hamiltonian graph*.

The *k-Hamiltonian graphs* have been studied in many literatures. Some previous results [1, 2, 6, 10, 13, 14] were focused on the case for  $k = 1$ . In [14], Wang et al. presented a construction scheme, called 3-join. Many other 3-regular 1-Hamiltonian graphs [1, 2, 6, 10, 13] can be constructed by 3-join. Some researchers studied the properties of *k-Hamiltonian graphs* for  $k \geq 1$ . Many graphs have been showed to be *k-Hamiltonian graphs*, including recursive circulant graphs [12], crossed cubes [3], twisted cubes [5], Möbius cubes [4], pancake graphs [8]. In [7], Hung et al. presented a construction scheme, called node expansion. A family of *k-Hamiltonian graphs* can be constructed with node expansion, for  $k \geq 1$ . In this paper, we propose a construction scheme, called  $(k + 2)$ -join, for strongly *k-Hamiltonian graphs*. The  $(k + 2)$ -join is the generation of both 3-join [14] and node expansion [7]. Additionally, we present another construction scheme by Cartesian product for strongly *k-Hamiltonian graphs*.

The rest of this paper is organized as follows. In Section 2, we introduce the  $(k + 2)$ -join

operation. The construction scheme by Cartesian product is presented in Section 3. Section 4 includes the concluding remark.

## 2 Strongly $k$ -Hamiltonian Graphs and $(k + 2)$ -Join Operation

In this section, we will introduce the  $(k + 2)$ -join operation and show that the  $(k + 2)$ -join of two  $(k + 2)$ -regular and strongly  $k$ -Hamiltonian graphs is also strongly  $k$ -Hamiltonian.

Let  $x$  be a node of the  $(k + 2)$ -regular graph  $G = (V_g, E_g)$  and  $y$  be a node of the  $(k + 2)$ -regular graph  $H = (V_h, E_h)$  with  $V_g \cap V_h = \emptyset$ . Let  $N(x) = \{x_1, x_2, \dots, x_{k+2}\}$  be the neighbor of  $x$  and  $N(y) = \{y_1, y_2, \dots, y_{k+2}\}$  be the neighbor of  $y$ . The  $(k + 2)$ -join of  $G$  and  $H$ , denoted by  $G \sqcup H$ , at  $x$  and  $y$  is described as follows.

$$V(G \sqcup H) = V_g \cup V_h - \{x, y\}, \text{ and}$$

$$E(G \sqcup H) = E_g \cup E_h \cup \{(x_i, y_i) | 1 \leq i \leq k + 2\} - (\{(x, x_i) | x_i \in N(x)\} \cup \{(y, y_i) | y_i \in N(y)\}).$$

The graphs  $(G \sqcup H)$ ,  $G$  and  $H$  are illustrated in Figure 1. The  $(k + 2)$ -join of two  $(k + 2)$ -regular graphs is still  $(k + 2)$ -regular.

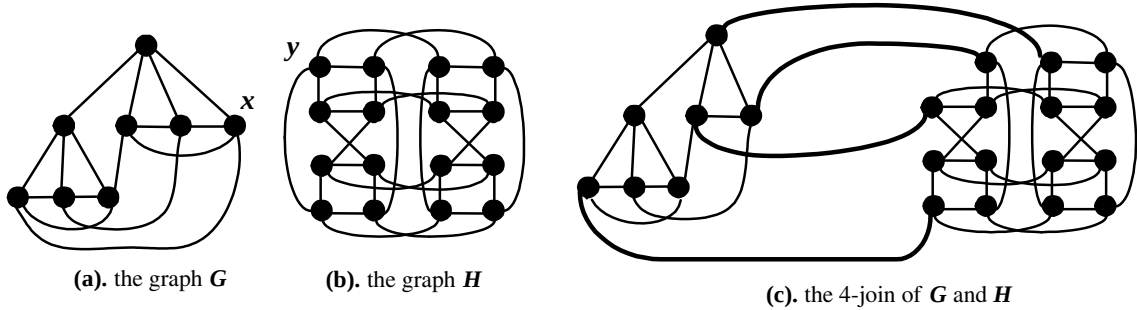


Figure 1: The  $(k + 2)$ -join of  $G$  and  $H$

The operation 3-join proposed in [14] is the special case of  $(k + 2)$ -join with  $k = 1$ . And the

operation node expansion, defined in [7], is the special case of  $(k + 2)$ -join where the graph  $H$  is a complete graph  $K_{k+3}$ . Thus the graph constructed with 3-join or node expansion can be constructed with  $(k + 2)$ -join.

In the follows, we will show the complete graph  $K_n$  is a strongly  $(n - 3)$ -Hamiltonian graph.

Let  $K_n = (V, E)$  be a complete graph. The following theorem is proved by Ore [11].

**Theorem 1** [11] *For  $F \subset E$ , the graph  $K_n - F$  is Hamiltonian if  $|F| \leq n - 3$ , and Hamiltonian connected if  $|F| \leq n - 4$ .*

A graph  $G$  is said to be  $r$ -HP if there exists  $V' \subseteq V(G)$  with  $|V'| = r$  such that every pair of nodes in  $V'$  can be joined by a Hamiltonian path of  $G$ . The following theorem is showed in [7].

**Theorem 2** [7] *The graph  $K_n - F$  is  $(n - f)$ -HP for every  $F \subset (V \cup E)$  with  $|F| = f \leq n - 2$ , where  $K_n = (V, E)$ .*

Applying **Theorem 1** and **Theorem 2**, we can obtain the following lemma.

**Lemma 1** *The complete graph  $K_n$  is strongly  $(n - 3)$ -Hamiltonian for  $n \geq 4$ .*

**Proof.** We will prove this lemma by induction on  $n$ . Since  $K_4$  is 3-regular and 1-Hamiltonian,  $K_4$  is strongly 1-Hamiltonian. Assume the complete graph  $K_t$  is strongly  $(t - 3)$ -Hamiltonian for  $4 \leq t \leq n - 1$ .

Let  $F$  be the faults of  $K_n = (V, E)$  with  $|F| = f \leq n - 3$ . First, we consider that  $|F \cap V| = i > 0$ . Thus the graph  $K_n - F$  is isomorphic to  $K_{n-i} - F'$  for some  $F' \subset E$  and  $|F'| \leq f - i$ . By induction hypotheses, the graph  $K_{n-i}$  is strongly  $(n - i - 3)$ -Hamiltonian.

Hence,  $\forall v \in (V(K_{n-i}) - F')$  there exist  $n - i - 1 - |F'| \geq n - 1 - f$  edges incident to  $v$  such that every pair of these edges is on some Hamiltonian cycle of  $K_{n-i} - F'$  for all  $F' \subseteq E$  and  $|F'| \leq n - i - 3$ . Therefore,  $\forall v \in (V(K_n) - F)$  there exist  $n - 1 - f$  edges incident to  $v$  such that every pair of these edges is on some Hamiltonian cycle of  $K_n - F$  for all  $F \subseteq (V \cup E)$  and  $|F| \leq n - 3$ . Thus,  $K_n$  is strongly  $(n - 3)$ -Hamiltonian.

Next, we consider that  $F \subset E$ . Applying **Theorem 1**, we can obtain that when  $f = n - 3$ ,  $K_n - F$  is Hamiltonian. This satisfies the conditions of strongly  $(n - 3)$ -Hamiltonian graphs with  $f = n - 3$ . Now we consider that  $f \leq n - 4$ . Let  $E(v)$  denote the set of edges incident to  $v$  for  $v \in V$ . Since  $f \leq n - 4$ , there exists a node  $u$  such that  $|E(u) \cap F| \leq 1$ . Let  $G_u$  be the subgraph of  $K_n$  induced by  $V - \{u\}$  and  $F_{G_u} = F \cap (V(G_u) \cup E(G_u))$ . Thus  $G_u$  is isomorphic to  $K_{n-1}$ . By induction hypotheses,  $G_u$  is strongly  $(n - 4)$ -Hamiltonian. We will show  $K_n$  is strongly  $(n - 3)$ -Hamiltonian for  $f \leq n - 4$  by the following cases.

**Case 1:**  $|E(u) \cap F| = 1$ .

Thus  $|F_{G_u}| = |F| - 1 \leq n - 5$ . Since  $G_u$  is strongly  $(n - 4)$ -Hamiltonian,  $\forall v \in (V(G_u) - F_{G_u})$  there exist  $n - 2 - |F_{G_u}| = n - 1 - |F|$  edges incident to  $v$  such that every pair of these edges is on some Hamiltonian cycle of  $G_u - F_{G_u}$ . These Hamiltonian cycles can be extended to Hamiltonian cycles of  $K_n - F$  since  $|E(u) \cap F| = 1$ . Applying **Theorem 1**, we can obtain that  $G_u - F_{G_u}$  is Hamiltonian connected. Hence, every pair of  $E(u) - F$  is on some Hamiltonian cycle of  $K_n - F$ . Therefore,  $\forall v \in (V - F)$  there exist  $n - 1 - |F|$  edges incident to  $v$  such that every pair of these edges is on some Hamiltonian cycle of  $K_n - F$ . Thus  $K_n$  is strongly  $(n - 3)$ -Hamiltonian.

**Case 2:**  $|E(u) \cap F| = 0$ .

Thus  $|F_{G_u}| = |F| \leq n - 4$ . Since  $G_u$  is strongly  $(n - 4)$ -Hamiltonian,  $\forall v \in (V(G_u) - F_{G_u})$  there exist  $n - 2 - |F_{G_u}| = n - 2 - |F|$  edges incident to  $v$ , denoted by  $E_v$ , such that every pair of  $E_v$  is on some Hamiltonian cycle of  $G_u - F_{G_u}$ . These Hamiltonian cycles can be extended to

Hamiltonian cycles of  $K_n - F$  since  $|E(u) \cap F| = 0$ . Moreover,  $\forall v \in (V(G_u) - F_{G_u})$  every pair of  $E_v \cup \{(u, v)\}$  is on some Hamiltonian cycle of  $K_n - F$ . Applying **Theorem 2**, we can obtain that  $G_u - F_{G_u}$  is  $(n - 4 - |F_{G_u}|)$ -HP. Hence, there exist  $n - 1 - |F_{G_u}| = n - 1 - |F|$  nodes of  $V(G_u)$  which can be joined by a Hamiltonian path of  $G_u - F_{G_u}$ . Thus there exist  $n - 1 - |F|$  edges incident to  $u$  such that every pair of these edges is on some Hamiltonian cycle of  $K_n - F$ . Therefore  $K_n$  is strongly  $(n - 3)$ -Hamiltonian.

This lemma is follows. □

In the following theorem, we will show that the  $(k + 2)$ -join of two  $(k + 2)$ -regular and strongly  $k$ -Hamiltonian graphs is also strongly  $k$ -Hamiltonian.

Let  $F$ ,  $F_g$  and  $F_h$  denote sets of faulty components including faulty nodes and edges in  $G \sqcup H$ ,  $G$  and  $H$ , respectively. And let  $F_{gh} = F \cap \{(x_i, y_i) | 1 \leq i \leq k + 2\}$ . Thus  $F = F_g \cup F_h \cup F_{gh}$ .

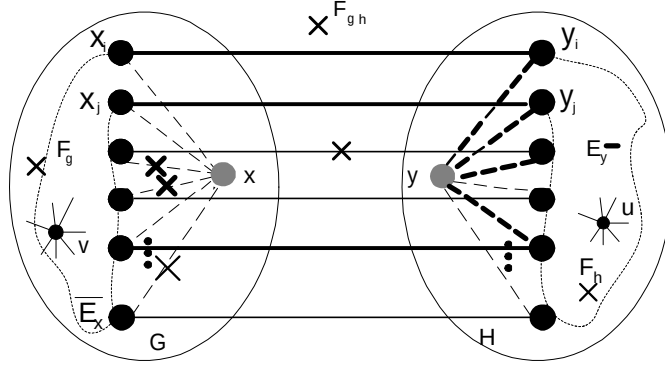


Figure 2: The strongly  $k$ -Hamiltonian property of  $G \sqcup H$ .

**Theorem 3** *If  $G$  and  $H$  are  $(k + 2)$ -regular and strongly  $k$ -Hamiltonian graphs, then  $G \sqcup H$  is also a strongly  $k$ -Hamiltonian graph.*

**Proof.** We will prove that  $\forall v \in (V(G \sqcup H) - F)$ , there exist  $(k + 2) - |F|$  edges incident to

$v$  (denoted by  $E_v$ ), such that every pair of  $E_v$  is on some Hamiltonian cycle of  $(G \sqcup H) - F$ ,  $|F| \leq k$ . Let  $G \sqcup H$  be the  $(k+2)$ -join of  $G$  and  $H$  at  $x$  and  $y$ , respectively. Thus, let  $\overline{E_x} = \{(x, x_i) | (y, y_i) \notin E_y, \text{ or } (x_i, y_i) \in F_{gh}\}$ . Hence  $|\overline{E_x}| \leq |F_h| + |F_{gh}|$ . We delete every edge in  $\overline{E_x}$  of  $G$ . Thus the faults of  $G$  is at most  $|F_g| + |\overline{E_x}| \leq |F_g| + |F_h| + |F_{gh}| = |F|$ . Since  $G$  is strongly  $k$ -Hamiltonian,  $\forall v \in (V_g - F_g)$  there exist  $k+2-|F|$  edges incident to  $v$  such that every pair of these edges is on some Hamiltonian cycle of  $G - F_g - \overline{E_x}$ . Let  $HC_g$  be an arbitrary Hamiltonian cycle of  $G - F_g - \overline{E_x}$ . Thus,  $HC_g$  contains two edges incident to  $x$ ,  $(x, x_i)$  and  $(x, x_j)$  for some  $y_i, y_j \in E_y$ . Since  $(y, y_i), (y, y_j) \in E_y$ , there exists a Hamiltonian cycle  $HC_h$  of  $H - F_h$  which contains  $(y, y_i)$  and  $(y, y_j)$ . Hence,  $HC_g \cup HC_h - \{x, y\} \cup \{(x_i, y_i), (x_j, y_j)\}$  forms a Hamiltonian cycle of  $(G \sqcup H) - F - \overline{E_x}$ . This construction is illustrated in Figure 2. Thus,  $\forall v \in (V_g - F_g)$  there exist  $k+2-|F|$  edges incident to  $v$  such that every pair of these edges is on some Hamiltonian cycle of  $(G \sqcup H) - F$ . With the similar construction,  $\forall u \in (V_h - F_h)$ , there exist  $k+2-|F|$  edges incident to  $u$  such that every pair of these edges is on some Hamiltonian cycle of  $(G \sqcup H) - F$ . Therefore,  $G \sqcup H$  is strongly  $k$ -Hamiltonian.  $\square$

Applying **Lemma 1** and **Theorem 3**, we can obtain that the graphs constructed in [7] are all strongly  $k$ -Hamiltonian. On the other hand, the graphs constructed only by **Lemma 1** and **Theorem 3** also can be constructed by node expansion in [7]. We need some other construction schemes to produce new strongly  $k$ -Hamiltonian graphs.

### 3 Strongly $k$ -Hamiltonian Graphs and Cartesian Product Operation

In this section, we will show that the Cartesian product of a  $(k+2)$ -regular, strongly  $k$ -Hamiltonian graph and  $K_2$  is strongly  $(k+1)$ -Hamiltonian.

The graph  $G \times H$  is the Cartesian product of the graphs  $G = (V_g, E_g)$  and  $H = (V_h, E_h)$ . The

node set  $V(G \times H)$  is  $\{(u, v) | \forall u \in V_g \text{ and } v \in V_h\}$ . The edge set  $E(G \times H)$  is  $\{((u_1, v_1), (u_2, v_2)) | \text{if } ((u_1, u_2) \in E_g \text{ and } v_1 = v_2) \text{ or } (u_1 = u_2, \text{ and } (v_1, v_2) \in E_h)\}$ . Thus,  $V = V(G \times K_2) = \{(v, i) | \forall v \in V(G) \text{ and } i = 1, 2\}$  and  $E = E(G \times K_2) = \{((u, i), (v, j)) | \text{if } ((u, v) \in E(G) \text{ and } i = j) \text{ or } u = v \text{ and } i \neq j\}$ . Let  $G_i$  be a sub-graph of  $(G \times K_2)$  induced by vertex set  $\{(v, i)\}$  for  $i = 1, 2$ . To be specific, the node set of  $G_i$  is  $\{(v, i) | \forall v \in V\}$  and the edge set of  $G_i$  is  $\{((u, i), (v, i)) | \forall (u, v) \in E\}$ .

Let  $F$ ,  $F_1$  and  $F_2$  denote the sets of faulty components including faulty nodes and edges in  $(G \times K_2)$ ,  $G_1$  and  $G_2$ , respectively. Let  $F_3$  be the set of faulty edges between  $G_1$  and  $G_2$ . That is,  $F_3 = F \cap \{((u, 1), (u, 2)) | \forall u \in V\}$ . Thus  $F = F_1 \cup F_2 \cup F_3$ .

**Theorem 4** *If  $G$  is  $(k + 2)$ -regular and strongly  $k$ -Hamiltonian,  $G \times K_2$  is strongly  $(k + 1)$ -Hamiltonian.*

**Proof.** We will prove that for all  $(v, i) \in (V - F)$ , there are  $k + 3 - |F_1|$  edges incident to  $(v, i)$  such that every pair of these edges is on some Hamiltonian cycle of  $G \times K_2 - F$ . And we will show this theorem with the following cases.

**Case 1:**  $|F| = k + 1$

Since  $G$  is strongly  $k$ -Hamiltonian, it is also  $k$ -Hamiltonian. In [9] C.N. Hung proved that if  $G$  is  $(k + 2)$ -regular and  $k$ -Hamiltonian then  $G \times K_2$  is  $(k + 1)$ -Hamiltonian. Thus, there exists a Hamiltonian cycle in  $G \times K_2 - F$ . That is,  $\forall (v, i) \in (V - F)$  there exist  $2 = (k + 3) - |F|$  edges incident to  $(v, i)$  such that these two edges are on a Hamiltonian cycle of  $G \times K_2 - F$ . Therefore,  $G \times K_2$  is strongly  $(k + 1)$ -Hamiltonian when  $|F| = k + 1$ .

**Case 2:**  $|F| \leq k$  and  $(F = F_1 \text{ or } F = F_2)$ . Without loss of generality, we can assume that  $F = F_1$ .



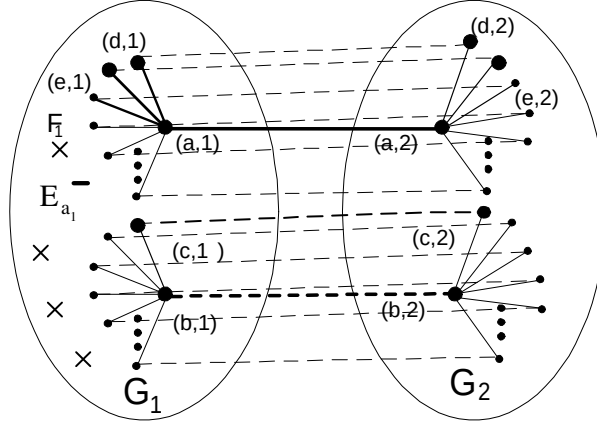


Figure 3: There exist  $k + 3 - |F|$  edges incident to  $(a, 1)$  such that every pair of these edges is on some Hamiltonian cycle of  $G \times K_2 - F$  if  $F = F_1$ .

**Case 2.1:** Consider the nodes of  $G_1$ .

Let  $(a, 1)$  be an arbitrary node of  $G_1$ . Since  $G_1$  is strongly  $k$ -Hamiltonian, there are  $k + 2 - |F_1|$  edges incident to  $(a, 1)$ , denoted by  $E_{a_1}$ , and every pair of  $E_{a_1}$  is on some Hamiltonian cycle of  $G_1 - F_1$ . Let  $HC_1$  be an arbitrary Hamiltonian cycle of  $G_1 - F_1$  contains two edges of  $E_{a_1}$ . Let  $(b, 1) \neq (a, 1)$  be a fault-free node. Thus,  $HC_1$  contains two edges incident to  $(b, 1)$ . At least one edge of these two edges denoted by  $((b, 1), (c, 1))$ , is not incident to  $(a, 1)$ . Since  $|F_2| = 0$ , the edge  $((b, 2), (c, 2))$  is on some Hamiltonian cycle  $HC_2$  of  $G_2$ . Thus,  $HC_1 \cup HC_2 - \{((b, 1), (c, 1)), ((b, 2), (c, 2))\} \cup \{((b, 1), (b, 2)), ((c, 1), (c, 2))\}$  forms a Hamiltonian cycle of  $G \times K_2 - F$ . Hence, there exist  $|E_{a_1}|$  edges incident to an arbitrary node  $(a, 1) \in V(G_1)$  such that every pair of  $E_{a_1}$  is on some Hamiltonian cycle of  $G \times K_2 - F$ . Additionally,  $HC_1$  also contains two edges incident to  $(a, 1)$ , denoted by  $((a, 1), (d, 1))$  and  $((a, 1), (e, 1))$ . The edges  $((a, 2), (d, 2))$  and  $((a, 2), (e, 2))$  are on some Hamiltonian cycle  $HC_3$  and  $HC_4$ , respectively of  $G_2$ . Thus,  $HC_1 \cup HC_3 - \{((a, 1), (d, 1)), ((a, 2), (d, 2))\} \cup \{((a, 1), (a, 2)), ((d, 1), (d, 2))\}$  and  $HC_1 \cup HC_4 - \{((a, 1), (e, 1)), ((a, 2), (e, 2))\} \cup \{((a, 1), (a, 2)), ((e, 1), (e, 2))\}$  are both Hamiltonian cycles of  $G \times K_2 - F$ . Figure 3 is illustrated for this case. Hence, every pair of

$E_{a_1} \cup \{((a, 1), (a, 2))\}$  is on some Hamiltonian cycle of  $G \times K_2 - F$ . Therefore,  $\forall(a, 1) \in V_1 - F$ , there exist  $k + 3 - |F|$  edges incident to  $(a, 1)$  such that every pair of these edges is on some Hamiltonian cycle of  $G \times K_2 - F$ .

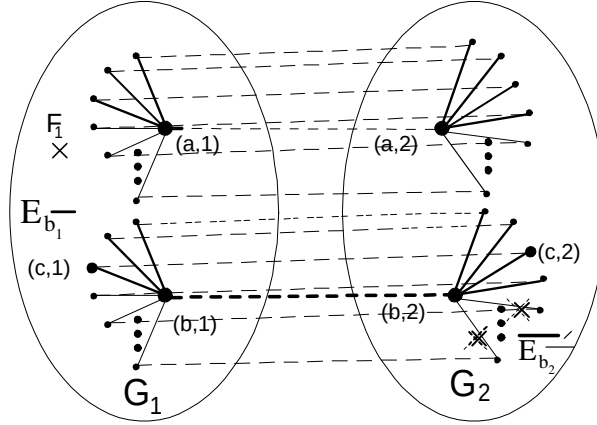


Figure 4: There exist  $k + 3 - |F|$  edges incident to  $(a, 2)$  such that every pair of these edges is on some Hamiltonian cycle of  $G \times K_2 - F$  if  $F = F_1$ .

**Case 2.2:** Consider the nodes in  $G_2$ .

Let  $(a, 2)$  be an arbitrary node of  $G_2$ . Let  $(b, 1)$  be a fault-free node of  $G_1$  and  $(b, 2) \neq (a, 2)$ . Since  $G_1$  is strongly  $k$ -Hamiltonian, there exist  $k + 2 - |F_1|$  edges incident to  $(b, 1)$  (denoted by  $E_{b_1}$ ) such that every pair of  $E_{b_1}$  is on some Hamiltonian cycle of  $G_1 - F_1$ . Let  $\overline{E_{b_2}}$  be the edge set  $\{((b, 2), (v, 2)) \mid (v, 1) \in V(G_1) \text{ and } ((b, 1), (v, 1)) \notin E_{b_1}\}$ . Thus,  $|\overline{E_{b_2}}| = |F_1|$ . We delete  $|\overline{E_{b_2}}| - 1$  edges of  $\overline{E_{b_2}}$  (denoted by  $\overline{E_{b_2}}'$ ). Since  $G_2$  is strongly  $k$ -Hamiltonian, there exist  $k + 2 - (|\overline{E_{b_2}}| - 1) = k + 3 - |F|$  edges incident to  $(a, 2)$  such that every pair of these edges is on some Hamiltonian cycle  $HC_2$  of  $G_2 - \overline{E_{b_2}}'$ . Thus,  $HC_2$  contains at least one edge  $((b, 2), (c, 2))$  incident to  $(b, 2)$  where  $((b, 1), (c, 1)) \in E_{b_1}$ . Since  $((b, 1), (c, 1)) \in E_{b_1}$ ,  $((b, 1), (c, 1))$  is on some Hamiltonian cycle  $HC_1$  of  $G_1 - F_1$ . Hence,  $HC_1 \cup HC_2 - \{((b, 1), (c, 1)), ((b, 2), (c, 2))\} \cup \{((b, 1), (b, 2)), ((c, 1), (c, 2))\}$  forms a Hamiltonian cycle of  $G \times K_2 - F$ . Figure 4 is illustrated for this case. Therefore,  $\forall(a, 2) \in V(G_2)$  there exist  $k + 3 - |F|$  edges incident to  $(a, 2)$  such

that every pair of these edges is on some Hamiltonian cycle of  $G \times K_2 - F$ .

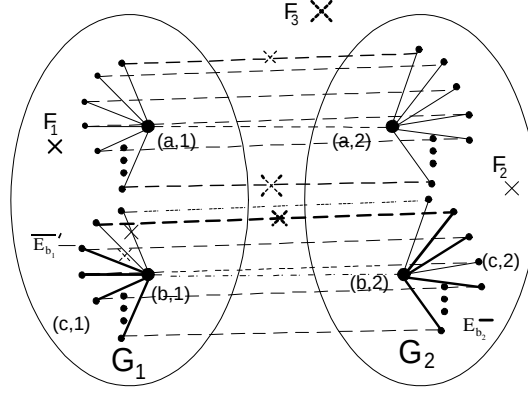


Figure 5: The case for  $|F| > |F_1|$ ,  $|F| > |F_2|$  and  $|F| \leq k$  in  $G \times K_2$

**Case 3:**  $|F| > |F_1|$  and  $|F| > |F_2|$  and  $|F| \leq k$ .

Let  $(a, 1), (b, 1), (b, 2)$  be fault-free nodes and  $((b, 1), (b, 2))$  be a fault-free edge. Since  $G_2$  is strongly  $k$ -Hamiltonian, there are  $k + 2 - |F_2|$  edges incident to  $(b, 2)$  (denoted by  $E_{b_2}$ ) such that every pair of  $E_{b_2}$  is on some Hamiltonian cycle of  $G_2 - F_2$ . Let  $\overline{E_{b_1}}$  be the edge set  $\{((b, 1), (v, 1)) \mid \text{for } (v, 1) \in V(G_1) \text{ and } (((b, 2), (v, 2)) \notin E_{b_2} \text{ or } ((v, 1), (v, 2)) \in F_3)\}$ . Thus,  $|\overline{E_{b_1}}| \leq |F_2| + |F_3|$ . We delete  $|\overline{E_{b_1}}| - 1$  edges of  $\overline{E_{b_1}}$  (denoted by  $\overline{E_{b_1}}'$ ). Since  $G_1$  is strongly  $k$ -Hamiltonian,  $\forall (a, 1) \in V(G_1)$  there are  $k + 2 - |F_1| - |\overline{E_{b_1}}'| \geq k + 3 - |F|$  edges incident to  $(a, 1)$  such that every pair of these edges is on some Hamiltonian cycle of  $G_1 - F_1 - \overline{E_{b_1}}'$ . Let  $HC_1$  be an arbitrary Hamiltonian cycle contains a pair of these  $k + 3 - |F|$  edges incident to  $(a, 1)$  of  $G_1 - F_1 - \overline{E_{b_1}}'$ . There exists a edge  $((b, 1), (c, 1))$  of  $HC_1$  where  $((b, 2), (c, 2)) \in E_{b_2}$ . Since the edge  $((b, 2), (c, 2)) \in E_{b_2}$ , it is on some Hamiltonian cycle  $HC_2$  of  $G_2 - F_2$ . Hence,  $HC_1 \cup HC_2 - \{((b, 1), (c, 1)), ((b, 2), (c, 2))\} \cup \{((b, 1), (b, 2)), ((c, 1), (c, 2))\}$  forms a Hamiltonian cycle of  $G \times K_2 - F$ . Thereby,  $\forall (a, 1) \in (V(G_1) - F_1)$  there are  $k + 3 - |F|$  edges incident to  $(a, 1)$  such that every pair of these edges is on some Hamiltonian cycle of  $G \times K_2 - F$ . Figure 5 is illustrated for this case. Applying the similar construction scheme, we can obtain

that  $\forall(a, 2) \in (V(G_2) - F_2)$  there are  $k + 3 - |F|$  edges incident to  $(a, 2)$  such that every pair of these edges is on some Hamiltonian cycle of  $G \times K_2 - F$ . Therefore,  $G \times K_2$  is strongly  $(k + 1)$ -Hamiltonian.  $\square$

Applying **Lemma 1**, **Theorem 3** and **Theorem 4**, we can construct many strongly  $k$ -Hamiltonian graphs. Moreover, some graphs are also new  $k$ -Hamiltonian graphs. The graphs  $K_4 \times K_2$ ,  $K_5$  and  $(K_4 \times K_2) \sqcup K_5$  are strongly 2-Hamiltonian as illustrated in Figure 6.

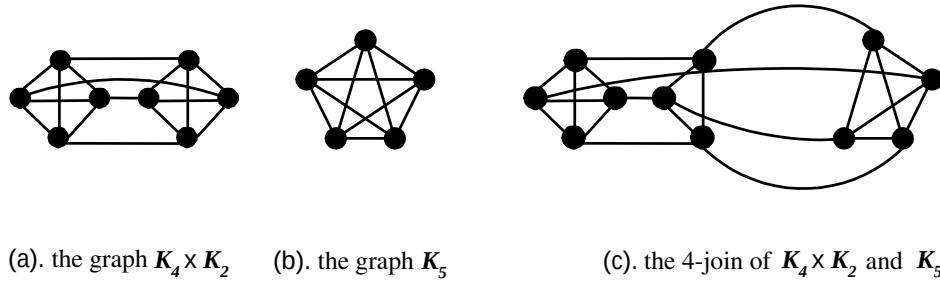


Figure 6: Some strongly  $k$ -Hamiltonian graphs

## 4 Conclusion

In this paper, we introduce the concept of strongly  $k$ -Hamiltonian graphs. We also present two construction schemes, the  $(k + 2)$ -join and Cartesian product with  $K_2$ , for strongly  $k$ -Hamiltonian graphs. Applying these schemes, we can construct a series of  $(k + 2)$ -regular and strongly  $k$ -Hamiltonian graphs. In the future, we hope to present more construction schemes for the strongly  $k$ -Hamiltonian graphs.

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